

KRAMFORS HOGA KUSTEN, SwedenWMO: **022590**Lat: **63.049N**Lon: **17.769E**Elev: **10**StdP: **101.20**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.2	-22.8	-28.1	0.3	-25.3	-25.0	0.4	-22.5	10.9	-0.1	9.6	-0.8	1.9	0	0.665

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	26.0	17.9	24.0	17.0	22.0	16.0	19.2	24.2	17.9	22.7	16.8	20.7	4.2	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	12.3	20.8	16.0	11.4	19.8	15.0	10.7	18.7	54.4	24.2	50.7	23.0	47.2	20.9	22.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	8.0	7.0	-26.7	28.2	4.5	2.6	-30.0	30.1	-32.6	31.6	-35.1	33.0	-38.4	34.9		
(5)			-27.4	19.7	4.3	1.8	-30.5	21.0	-33.1	22.0	-35.5	23.0	-38.6	24.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.9	-8.1	-7.9	-2.1	3.6	8.9	13.1	16.5	14.6	10.2	3.9	-1.3	-5.5
	DBStd	9.62	6.97	7.46	4.98	2.96	3.91	3.01	2.96	2.68	2.97	3.25	4.60	6.24
	HDD10.0	2749	560	502	374	193	69	6	0	1	33	191	340	479
	HDD18.3	5293	818	736	633	443	292	158	74	120	244	448	590	738
	CDD10.0	522	0	0	0	0	35	99	202	144	39	1	0	0
	CDD18.3	24	0	0	0	0	0	2	18	4	0	0	0	0
	CDH23.3	236	0	0	0	0	12	29	168	26	0	0	0	0
	CDH26.7	39	0	0	0	0	0	3	34	1	0	0	0	0
(14) Wind	WSAvg	2.8	2.6	2.5	3.1	2.9	3.2	3.5	3.0	2.7	2.6	2.6	2.5	2.4
(15) Precipitation	PrecAvg	621	46	29	30	35	40	54	89	80	59	57	53	49
	PrecMax	913	86	62	76	86	90	111	179	166	148	162	153	100
	PrecMin	433	5	5	6	6	1	22	10	22	19	12	21	7
	PrecStd	113	21	14	14	21	21	22	43	42	37	35	32	21
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.1	7.8	11.0	18.8	25.0	26.8	29.9	26.0	21.0	14.1	9.8	7.0
		MCWB	2.8	4.2	6.3	9.8	14.9	16.9	18.8	19.3	14.6	10.2	7.8	4.5
	2%	DB	3.1	6.0	8.1	15.1	22.0	23.8	27.2	23.9	18.8	12.0	6.9	4.9
		MCWB	2.1	3.4	4.1	8.6	13.8	15.9	18.7	17.3	14.0	9.5	6.2	3.2
	5%	DB	2.1	3.1	6.8	12.2	19.1	21.2	25.0	21.9	17.0	10.9	5.8	3.2
		MCWB	1.4	1.3	3.3	6.7	12.2	14.4	18.1	16.6	13.2	8.7	4.9	2.4
	10%	DB	1.0	1.2	4.8	10.1	16.9	19.2	23.0	20.0	15.8	9.1	4.8	2.1
		MCWB	0.6	0.6	1.9	5.2	10.9	13.4	17.1	15.3	12.7	7.7	4.1	1.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	4.6	6.5	10.4	16.0	18.2	20.8	20.3	16.2	11.5	8.1	4.6
		MCDB	4.0	7.1	10.2	17.8	22.0	24.0	25.0	23.8	18.8	13.3	9.2	5.8
	2%	WB	2.2	3.3	4.6	8.6	14.5	16.7	19.7	18.7	15.0	10.3	6.1	3.3
		MCDB	2.9	5.6	7.5	14.4	20.8	21.6	24.9	22.4	17.5	11.6	6.3	4.0
	5%	WB	1.2	1.7	3.2	7.0	12.9	15.6	18.6	17.2	14.1	8.9	4.8	2.3
		MCDB	1.7	2.5	5.9	11.8	18.2	19.9	24.0	20.6	16.3	10.2	5.1	2.9
	10%	WB	0.3	0.6	2.0	5.6	11.2	14.4	17.6	16.1	13.1	7.7	3.6	1.4
		MCDB	0.7	1.1	4.0	9.3	16.2	18.1	22.4	19.0	15.2	8.9	4.1	1.8
(35) Mean Daily Temperature Range		MDBR	7.1	8.9	9.6	10.2	11.0	9.8	10.2	9.4	8.9	7.1	5.4	6.6
	5% DB	MCDBR	7.0	8.6	10.8	15.3	15.2	13.0	13.0	12.3	10.8	9.3	6.3	7.3
		MCWBR	6.4	6.9	7.3	8.9	8.1	6.6	6.2	7.0	6.9	7.0	5.4	6.5
	5% WB	MCDBR	6.8	7.6	9.6	14.6	14.0	11.3	11.8	10.2	9.3	8.3	5.9	6.8
		MCWBR	6.4	6.4	6.8	8.8	7.7	6.4	6.0	6.2	6.6	6.9	5.4	6.2
(40) Clear-Sky Solar Irradiance	taub	0.233	0.293	0.317	0.357	0.357	0.353	0.369	0.360	0.328	0.312	0.375	0.348	
	taud	2.064	2.204	2.294	2.348	2.425	2.469	2.443	2.465	2.541	2.431	2.717	2.602	
	Ebn at Noon	399	640	781	825	859	870	841	817	781	632	376	228	
	Edh at Noon	31	62	84	99	99	97	97	88	67	51	28	18	
(44) All-Sky Solar Radiation	RadAvg	0.17	0.79	2.26	3.80	4.86	5.57	5.09	3.86	2.36	1.00	0.27	0.09	
	RadStd	0.02	0.13	0.25	0.33	0.52	0.37	0.53	0.33	0.27	0.17	0.05	0.01	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Regional (41 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-123	N/A	N/A	N/A